

Forest Composition, Biodiversity, and Wildfire Resilience: Analyzing Biodiversity and Wildfire Outcomes on Indigenous and Non-Indigenous Lands Across Oregon

DATA 510 Project Proposal

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1 Title and Team Metadata

Project title: Forest Composition, Biodiversity, and Wildfire Resilience: Analyzing Biodiversity and Wildfire Outcomes on Indigenous and Non-Indigenous Lands Across Oregon

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Peer Stakeholder POs: Siera Edwards, Brooke Proctor, Serenna Walter (*Shared Theme: Pacific Northwest forest and environmental health.*)

Studio Session: 1

Team Size: 2

2 Problem Framing and Research Questions

2.1 Background: Oregon, forest structures, and fire stewardship

Tribes in the Pacific Northwest have been stewarding forests for millennia. These practices were (and still are) interrupted by American colonialism, which in turn impacted forest health and worsened outcomes from wildfires (Gleason, 2025). While there is momentum toward co-stewardship, issues such as federal funding gaps between tribes and the Forest Service impede effective fire stewardship (Gleason, 2025). Before policies against fire were in place, frequent low-severity fires were an important part of Indigenous fire stewardship, and without them, forest structures have shifted—the PNW’s mature and old-growth forests are at increased risk of more severe wildfires (Lundeberg, 2026). Increasingly frequent and extreme wildfires are threatening Oregon residents, the majority of whom live west of the Cascades, which are at heightened risk of fires, and rely on Cascadian watersheds for water and electrical power (Coughlan et al, 2024).

2.2 Why this project matters to our team

Both team members have researched Indigenous data governance and Indigenous data sovereignty and the ethical issues associated with them, and are eager to continue exploring these issues through a new, practical lens. We have also been impacted by fires and smoke in California and Oregon, and are invested in reducing risks while promoting ecological stability.

2.3 Problem statement (consequential framing)

Problem: Wildfires continue to threaten and impact Oregon residents and forests, and are exacerbated by climate change and the presence of landscape fuels created by fire-exclusion policies (Lundeberg, 2026).

Consequence if unsolved: The continued loss of mature and old-growth forests would impact the wider Oregon ecological landscape, and unchecked wildfires will impact Oregon residents through loss of homes and potentially increasing mortality rates and fire severity increases.

What success would change (this term): A method to see continuing change in forest composition and wildfire severity through open-source data and answers to our research questions below that are easily interpretable for our stakeholders with less technical backgrounds.

2.4 Research questions

Primary research questions:

1. RQ1: To what extent are federally recognized Indigenous lands characterized by different forest composition and wildfire severity outcomes compared with nearby ecologically similar non-Indigenous lands across the Pacific Northwest and Western United States?
 - a. Sub-RQ1A: Are Indigenous lands associated with greater forest compositional diversity than nearby non-Indigenous lands?
 - b. Sub-RQ1B: Are Indigenous lands associated with lower wildfire severity after controlling for forest type, climate region, and fire size?
 - c. Sub-RQ1C: Which ecological features (tree diversity, understory diversity, keystone species presence) are most predictive of wildfire severity?

3 Impact and Stakeholders

Who benefits:

- Indigenous community leaders in the PNW
 - Those living on tribal lands and hoping to gain a better understanding of their wildfire risk
- PNW residents at high risk of wildfires

- Communities with compounding risks (i.e. social vulnerabilities, other environmental hazards)
- Forest management & fire prevention officials
 - Environmental hazard preparedness organizations

Impact (plausible, not hype): Our stakeholders will have a more comprehensive understanding of wildfire risks and forest composition across Oregon in reference to Indigenous lands. We will also create a pipeline using publicly available data that will continue to be updated that future researchers can reference for up-to-date analysis.

Shared Theme and peer PO influence: Our peer POs are also exploring *Pacific Northwest forest and environmental health*. They will help guide our metrics of forest composition and wildfire severity, as well as pushing us to ensure our results sufficiently benefit our stakeholders.

4 Data Sources and Access Plan

Source	Description	Access	Volume (est.)	Refresh	Instructor approval
LandMark	Locally defined boundaries of lands that are held or used by Indigenous Peoples and local communities worldwide	Public download (link 1)	Polygon shapefile (~1.5GB)	As information is received	Pending
Global Biodiversity Information Facility (GBIF)	Location-tied species occurrence records	Public download through occurrence search portal (link 2)	Table, approx. 270,000 records	As information is received	Pending
NIFC InFORM Fire Occurrence Data Records	Fire Occurrence Data Record (FODR) produced by the Interagency Fire Occurrence Reporting Modules (InFORM) application	Public download (link 3)	Table, approx. 1.2 million records	As incident record is certified	Pending

NIFC Interagency Wildland Fire Perimeter History (IFPH)	Fire history perimeter data of conglomerated agency perimeters	Public download (link 4)	Polygon shapefile (~670MB)	Updated annually	Pending
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Documented access URLs:

1. LandMark: <https://landmarkmap.org/data-methods/access-data>
2. GBIF:
[https://www.gbif.org/occurrence/search?country=US&has_geospatial_issue=false&taxon_key=2877951&taxon_key=3020559&taxon_key=2875008&taxon_key=3040183&taxon_key=3189834&taxon_key=3001068&taxon_key=2684241&taxon_key=2876099&year=2015,2026&geometry=POLYGON\(\(-126.97904%2029.79026,-112.4575%2029.79026,-112.4575%2050.21368,-126.97904%2050.21368,-126.97904%2029.79026\)\)&occurrence_status=present](https://www.gbif.org/occurrence/search?country=US&has_geospatial_issue=false&taxon_key=2877951&taxon_key=3020559&taxon_key=2875008&taxon_key=3040183&taxon_key=3189834&taxon_key=3001068&taxon_key=2684241&taxon_key=2876099&year=2015,2026&geometry=POLYGON((-126.97904%2029.79026,-112.4575%2029.79026,-112.4575%2050.21368,-126.97904%2050.21368,-126.97904%2029.79026))&occurrence_status=present)
3. InFORM Fire Occurrence Data Records:
<https://data-nifc.opendata.arcgis.com/datasets/nifc::inform-fire-occurrence-data-records/about>
4. Interagency Fire Perimeter History:
<https://data-nifc.opendata.arcgis.com/datasets/nifc::interagencyfireperimeterhistory-all-years-view/about>

Note: LandMark data in the US changes very slowly, as it is based on locally defined borders. We do not expect this information to change over the course of the term.

Note: GBIF has an API, but we have opted not to use it as the frequency of updates does not align with our other sources (particularly the IFPH, which is annual).

5 Data Engineering Plan

Repo bootstrapped from LucasCordova/ds3template:

- `data/external/`: all of our data is open source, and a lot of it is large, so we will provide instructions/download links instead of the actual files
- `data/interim/`: versions of shapefile data with polygons removed (only non-geometry columns for each row)
- `data/processed/`: if small enough, compiled data without polygons but including grid position (ready for modeling)

We will save copies of raw data locally rather than hosting it on GitHub because they don't need to be regularly updated.

Feature engineering: To simplify our analysis, we will construct a 5km x 5km grid system where each grid will have markers for fire year and whether it is Indigenous/tribal land. We will also construct forest structure features based on keystone plant species from the GBIF data and a fire severity feature from the NIFC data (Tallamy, 2024).

6 Planned Analysis

Descriptive Statistics: Establish baseline differences in biodiversity, wildfire frequency, and wildfire severity between Indigenous and non-Indigenous lands. Summary statistics including means, medians, standard deviations, interquartile ranges, and distributional visualizations will be used to characterize ecological conditions across landscape units. The baseline comparison for this analysis will be regional averages calculated across all landscape units in the study area. Evaluation will focus on metrics such as species richness, biodiversity indices, wildfire frequency, and wildfire severity. This stage will be considered successful in answering the research question if it establishes whether biodiversity and wildfire outcomes appear systematically different between Indigenous and non-Indigenous lands.

Inferential Statistics: Evaluate whether observed differences between Indigenous and non-Indigenous lands are statistically distinguishable from random variation. Welch's t-tests will be used when comparing means under unequal variance assumptions, while Mann-Whitney U tests will be used when distributional assumptions are not satisfied. These methods will be applied to biodiversity measures, wildfire severity metrics, and wildfire frequency indicators. The null hypothesis for each comparison will state that no difference exists between Indigenous and non-Indigenous lands. The alternative hypothesis will state that a difference exists. Evaluation will rely on p-values, confidence intervals, and effect size measurements such as Cohen's d and rank-biserial correlation. This stage will be considered successful if statistically significant differences are identified, or if evidence demonstrates that meaningful differences are unlikely to exist. In either case, the analysis will provide direct evidence regarding whether Indigenous and non-Indigenous lands differ with respect to biodiversity and wildfire outcomes.

Multiple Regression Modeling: Determine whether Indigenous land status remains associated with wildfire severity after accounting for ecological and geographic factors. Predictor variables will include Indigenous land designation, species richness, biodiversity indices, keystone species presence, forest type, and ecoregion. Wildfire severity will serve as the primary response variable. A baseline model will first be constructed using ecological predictors alone. A second model will then incorporate Indigenous land status to determine whether it provides additional explanatory power. Model evaluation will include adjusted R^2 , root mean squared error (RMSE), mean absolute error (MAE), coefficient significance tests, and confidence intervals. Validation procedures will include residual diagnostics, variance inflation factor assessments for

multicollinearity, and cross-validation where computationally feasible. Success will occur if the model identifies meaningful relationships that remain robust after controls are introduced.

Machine Learning: Identify ecological variables that most strongly predict wildfire severity and to evaluate whether nonlinear relationships exist that may not be captured by traditional statistical models. Random Forest and XGBoost algorithms will be trained using biodiversity indicators, keystone species presence, forest composition measures, Indigenous land status, and geographic characteristics. Random Forest models will be evaluated using RMSE, MAE, R^2 , and out-of-bag error estimates. XGBoost models will be evaluated using the same predictive metrics and compared against both Random Forest models and baseline mean-prediction models. Validation procedures will include train-test splits, cross-validation, and hyperparameter tuning. SHAP analyses will be conducted to improve interpretability by identifying the ecological features that contribute most strongly to model predictions. Success will be achieved if predictive performance exceeds baseline approaches and if feature importance results remain stable across multiple modeling techniques.

Answering the Research Questions: If it can quantify biodiversity and wildfire severity differences between Indigenous and non-Indigenous lands, determine whether those differences are statistically significant, evaluate whether they remain after accounting for ecological confounding variables, and identify which ecological characteristics most consistently predict wildfire outcomes.

7 Program Integration (Five Pillars)

Pillar	Plan
Data engineering	Build a reproducible geospatial data pipeline integrating LandMark Indigenous land boundaries, GBIF biodiversity records, and NIFC wildfire datasets. Design a standardized 5km × 5km spatial grid, perform spatial joins, engineer biodiversity and wildfire features, and create processed datasets suitable for statistical and machine learning analyses.
Analytics / ML	Apply descriptive statistics, hypothesis testing, regression modeling, Random Forests, and XGBoost models to investigate relationships between Indigenous land status, biodiversity indicators, and wildfire severity outcomes. Use SHAP analysis to interpret model behavior and identify the ecological variables most associated with wildfire resilience.
Visualizations and communication	Develop maps, dashboards, comparative visualizations, regression summaries, and feature importance plots to communicate findings to both technical and non-technical audiences. Visual outputs will emphasize interpretability and support stakeholder decision-making regarding wildfire preparedness and ecological stewardship.
Ethics and responsible use	Incorporate CARE and FAIR principles, acknowledge Indigenous data sovereignty, avoid deficit framing, avoid causal claims unsupported by observational data, and ensure findings are communicated in ways that respect tribal diversity and governance autonomy.
Statistics / research design	Use observational comparative research methods to compare Indigenous and ecologically similar non-Indigenous lands. Apply hypothesis testing, regression controls, validation procedures, and sensitivity analyses to evaluate whether observed relationships remain consistent after accounting for ecological and geographic confounding factors.

8 Ethics and Risk Notes

Privacy: All datasets used are publicly available and do not contain personally identifiable information (PII). The project focuses on ecological and geographic characteristics rather than individual-level data.

Consent: The project relies exclusively on publicly released datasets made available by government agencies and nonprofit organizations. No direct human subjects research will be conducted, and no new data collection from individuals is planned.

Fairness and Representation: Potential risks include treating tribes as homogeneous. We must recognize that Indigenous nations represent diverse cultures, governance systems, ecological conditions, and stewardship practices, and strive to avoid generalized conclusions about all tribes, report regional variation when possible, and frame findings as landscape-level observations rather than cultural conclusions. Another potential risk is deficit framing; the project may unintentionally frame Indigenous lands solely through wildfire outcomes. We will focus on ecological characteristics and stewardship outcomes, avoid ranking management systems, and highlight strengths and limitations of all land management contexts. Overstating causality is another potential risk. It is important to recognize that observed relationships may be influenced by unmeasured ecological or historical factors. Our mitigation efforts will include clearly describing the study as observational, using association-based language, and avoiding causal claims unsupported by experimental evidence.

Indigenous Data Sovereignty: This project recognizes CARE (Collective Benefit, Authority to Control, Responsibility, Ethics) principles. Public datasets do not fully represent Indigenous ecological knowledge, stewardship practices, or decision-making processes. Findings should not be interpreted as substitutes for tribal expertise or community knowledge.

Data Retention and Stewardship: Raw data will be stored locally. Processed datasets and reproducible code will be maintained in the project repository. No personally identifiable information will be retained. Data usage will comply with source licensing requirements.

Spatial Limitations: Indigenous boundaries may not fully represent historical stewardship areas and grid-cell aggregation may obscure local ecological variation.

Biodiversity Limitations: GBIF observations reflect sampling effort and reporting practices and species absence cannot be inferred from missing observations.

Wildfire Limitations: Fire severity proxies may not fully capture ecological impacts. Fire reporting practices may vary across regions and years.

Modeling Limitations: Observational data cannot establish causation. Important ecological variables may be unavailable or unmeasured.

9 Timeline and Milestones

Week	Milestone	Deliverables
Week 4	M1 – Project Proposal	Finalize proposal, stakeholder analysis, ethics plan, data engineering plan, and analysis strategy
Week 5	Data Acquisition	Download, document, and catalog all datasets; begin cleaning
Week 6	Data Preparation	Complete cleaning, design grid system, engineer biodiversity and wildfire features
Week 7	M2 – Data Summary	Submit data inventory, schema documentation, quality assessment, reproducibility plan, and ethics review
Week 8	Statistical Analysis	Conduct descriptive statistics and hypothesis testing; refine engineered features
Week 9	Machine Learning Analysis	Train Random Forest and XGBoost models; conduct validation and feature importance analysis
Week 10	M3 – Poster Draft	Produce preliminary visualizations and draft poster content
Week 11	Model Refinement	Incorporate feedback and perform additional analyses as needed
Week 12	M4 – Report Draft	Complete full report draft and revise poster materials
Week 13	Final Visualization Development	Finalize maps, dashboards, regression figures, and SHAP visualizations
Week 14	M5 – Final Deliverables	Submit final report, poster, presentation, and reproducible project materials

Project Risks: Larger-than-anticipated data cleaning effort. Geospatial processing and shapefile conversion challenges. Complexity of grid-cell design and implementation. Difficulty operationalizing wildfire severity metrics. Technical challenges associated with spatial visualization and mapping workflows.

10 DS3 Process and Tooling

GitHub repo URL: <https://github.com/Fiden-Supancich-McCord/data510-fire>

GitHub Projects board URL: <https://github.com/orgs/Fiden-Supancich-McCord/projects/1>

Discord category: Category #studio confirmed for team; peer project categories joined.

Charter.md: Vision, mission, context, success criteria, working agreements, response SLAs, and Dor/DoD are committed within the charter. The backlog is seeded with milestone-tagged PBIs.

Studio loop: Studio briefs are due on Sunday by midnight, studio critiques are due on Wednesday by midnight. All peer content is to be submitted in the correct section in the GitHub repository, as well as linked in the #studio channel on Discord and submitted to the group GoogleChat. Iteration reviews (sans stakeholder response log) is to be posted in the corresponding repo by 5pm on Saturday.

10.1 Appendix: References

Coughlan M. R., Johnston J. D., Derr K. M., Lewis D. G. and Johnson B. R. (2024). Pre-contact Indigenous fire stewardship: a research framework and application to a Pacific Northwest temperate rainforest. *Front. Environ. Archaeol.* 3:1347571. doi: 10.3389/fearc.2024.1347571.

Gleason, S. (2025). Tribes lead the way in forest resilience. *The Washington State Standard*. <https://washingtonstatestandard.com/2025/09/25/tribes-lead-the-way-in-forest-resilience/>.

Lundeberg, S. (2026). Study: Old-growth forests face highest wildfire risk in areas that once burned frequently. *Philomath News, Oregon State University*. <https://philomathnews.com/study-old-growth-forests-face-highest-wildfire-risk-in-areas-that-once-burned-frequently/>.

Tallamy, D. (2024). Keystone native plants: marine West coast forests – ecoregion 7. *National Wildlife Federation*. <https://www.nwf.org/-/media/Documents/PDFs/Garden-for-Wildlife/Keystone-Plants/NWF-GFW-keystone-plant-list-ecoregion-7-marine-west-coast-forest.pdf>.